Arm Training Dataset — Visual Preview

Jessica Orbbec dual cameras · class **arm** · 2026-10-16

This PDF shows what the current training set looks like: raw labeled frames from cam0 (workcell / furnace view) and cam1 (rack view), plus auto-label overlays (yellow box) and a few YOLO smoke detections. There is no separate “gripper-only” dataset — the gripper is part of the single **arm** class box.

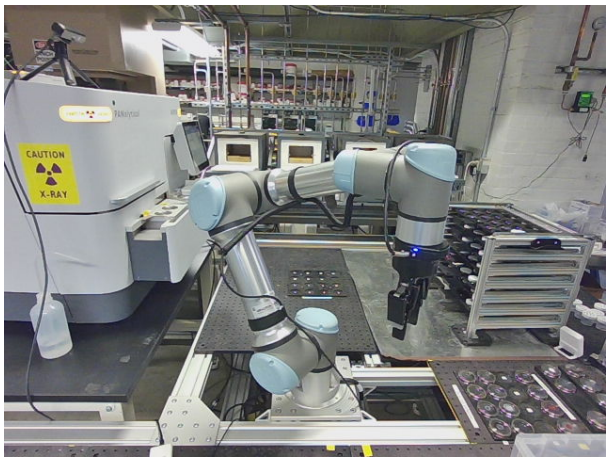
1. Dataset counts

Item	Count
captures / cam0	2476
captures / cam1	2474
keeps (labeled images)	4950
labels (.txt YOLO)	4950
dataset train	4208
dataset val	742
label method	color
class	arm

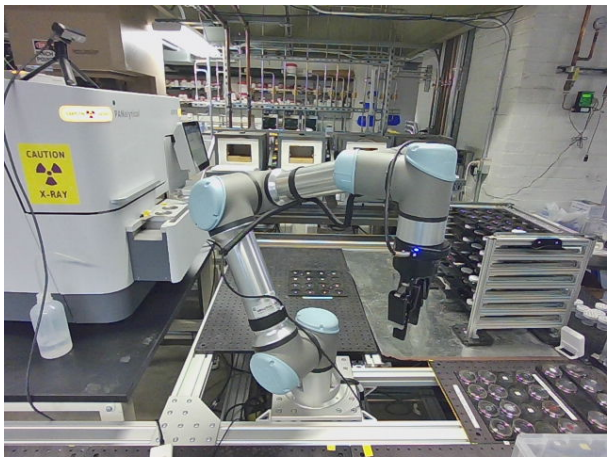
Weights: `C:\Users\Ceder-ALAB\Desktop\Jessica\Jessica\robotics\arm_state\runs\arm_detect\weights\best.pt`

2. Cam0 labeled frames (keeps)

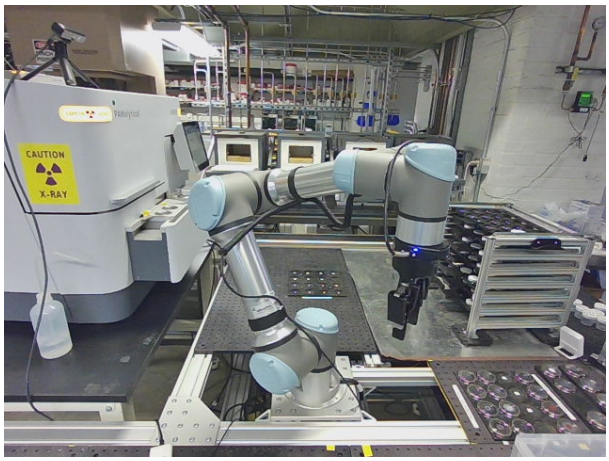
Overview camera — arm over the workcell. These are training images (evenly sampled across the keep set).



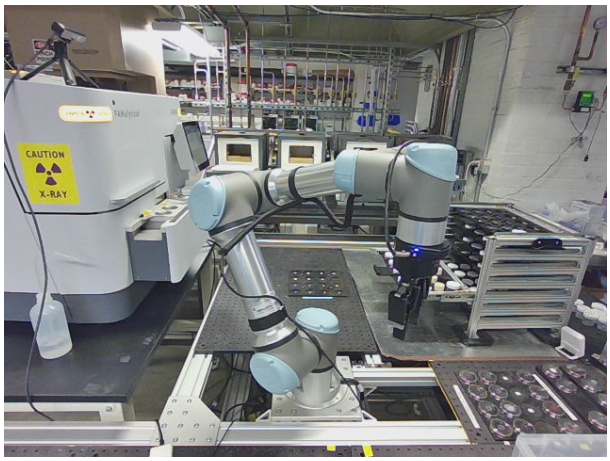
keep_cam0_01.png



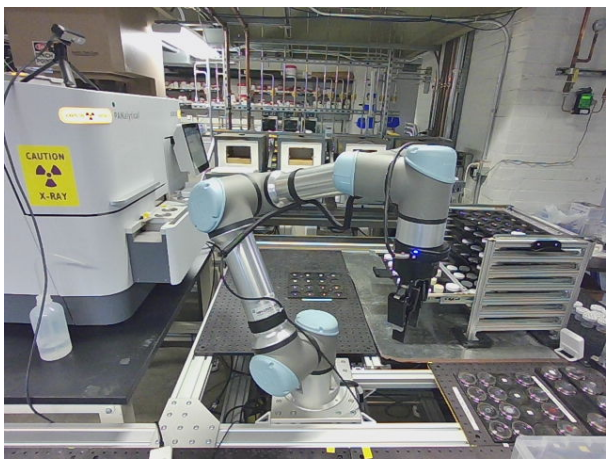
keep_cam0_02.png



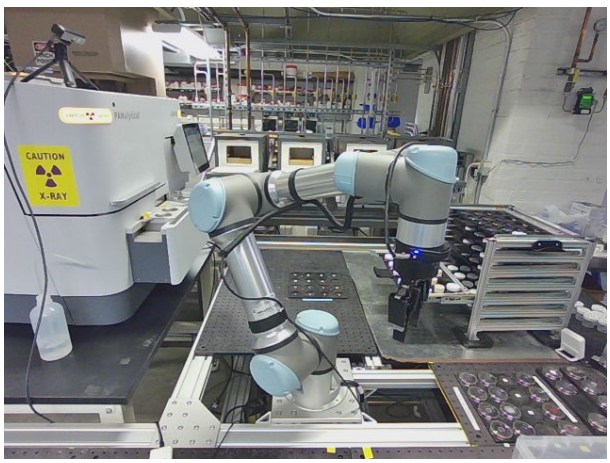
keep_cam0_03.png



keep_cam0_04.png



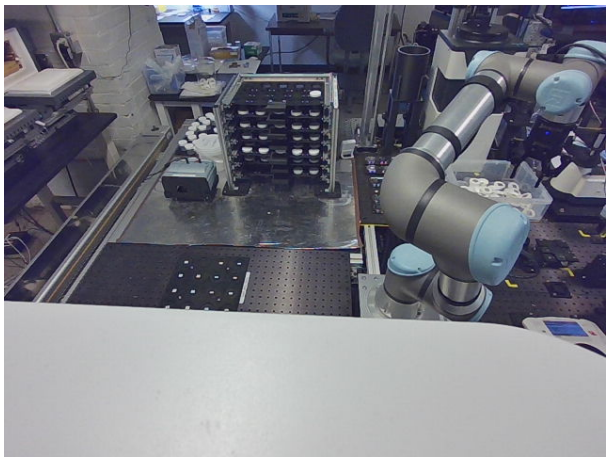
keep_cam0_05.png



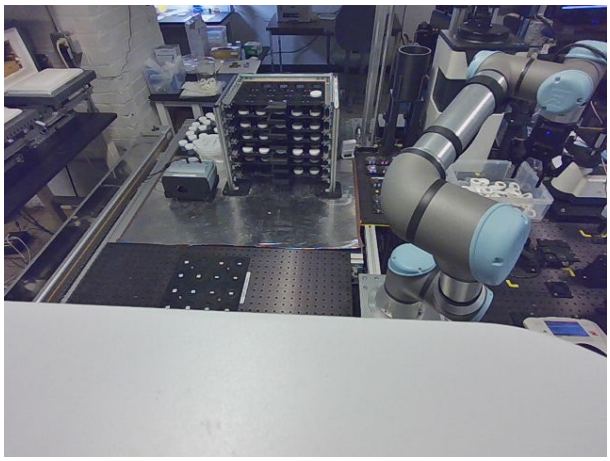
keep_cam0_06.png

3. Cam1 labeled frames (keeps)

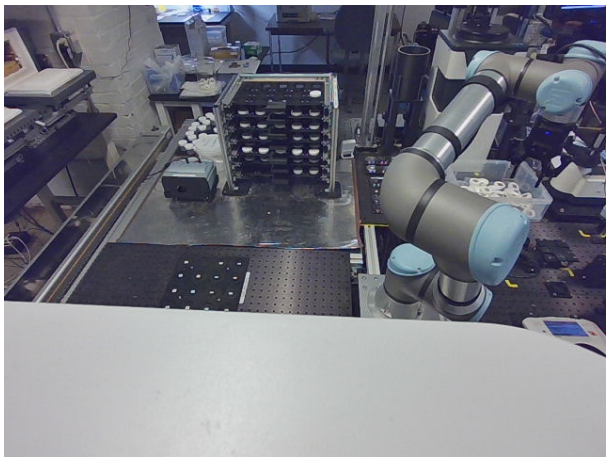
Rack-side camera — arm usually on the right of the frame.



keep_cam1_01.png



keep_cam1_02.png



keep_cam1_03.png



keep_cam1_04.png



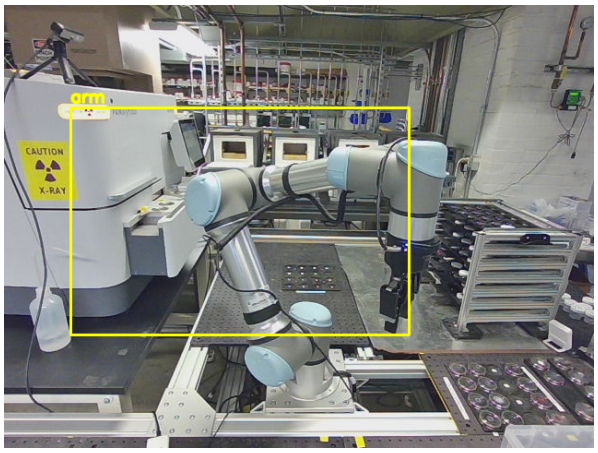
keep_cam1_05.png



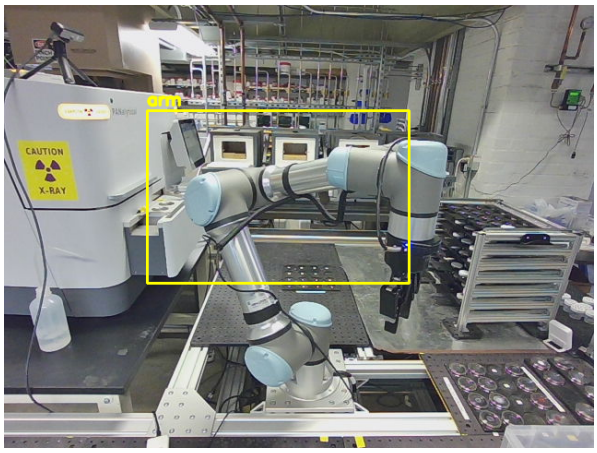
keep_cam1_06.png

4. Auto-label overlays (yellow “arm” box)

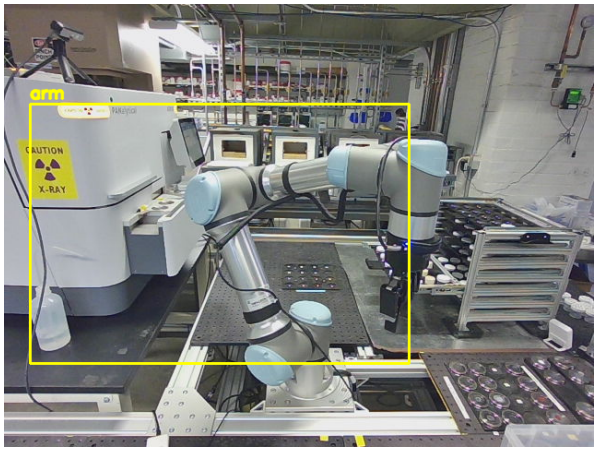
Boxes come from the UR cyan-joint color labeler (training labels), not from YOLO. Use these to judge label quality.



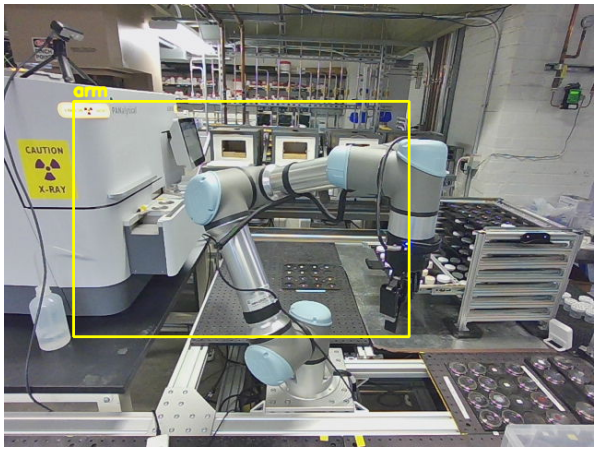
overlay_cam0_01.png



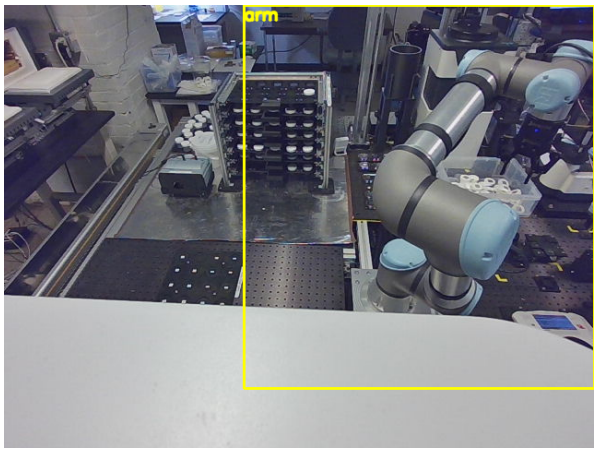
overlay_cam0_02.png



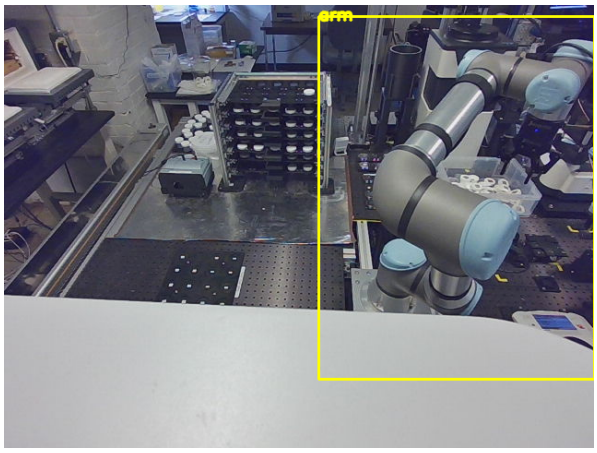
overlay_cam0_03.png



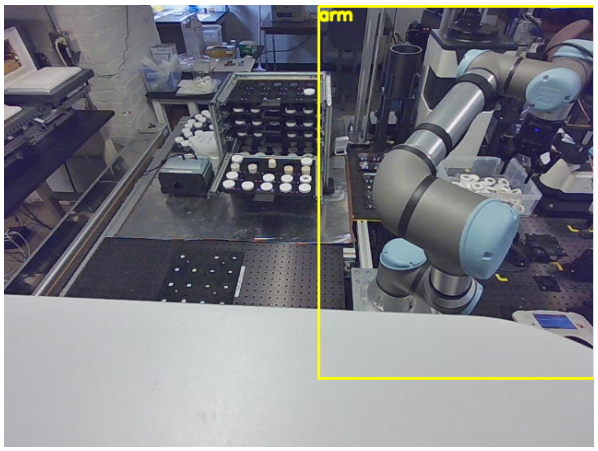
overlay_cam0_04.png



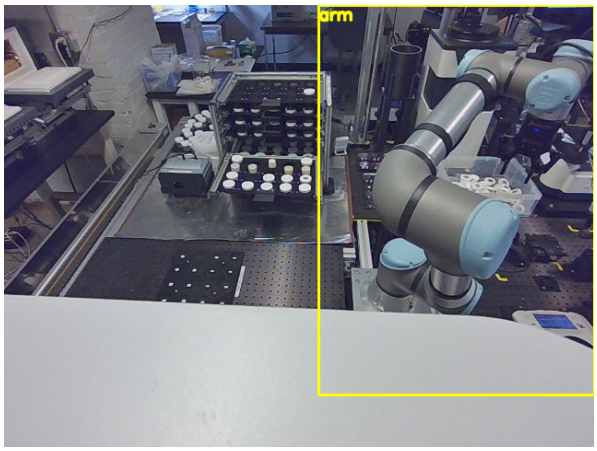
overlay_cam1_01.png



overlay_cam1_02.png



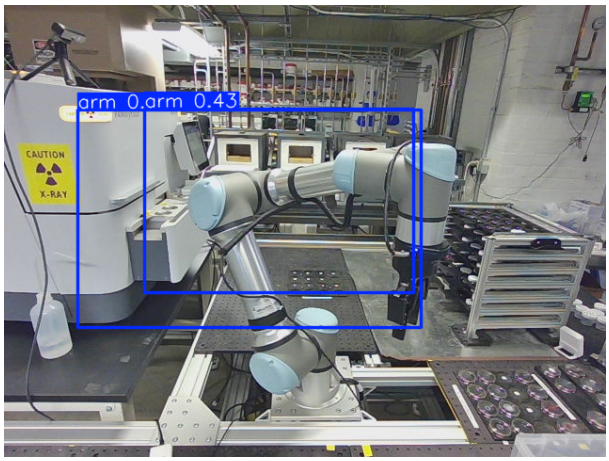
overlay_cam1_03.png



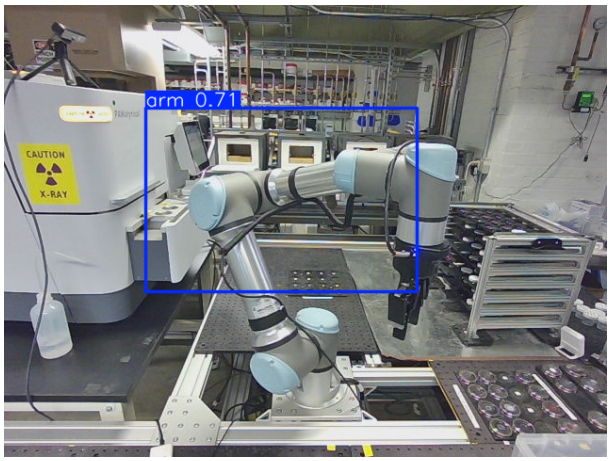
overlay_cam1_04.png

5. YOLO smoke detections (after train)

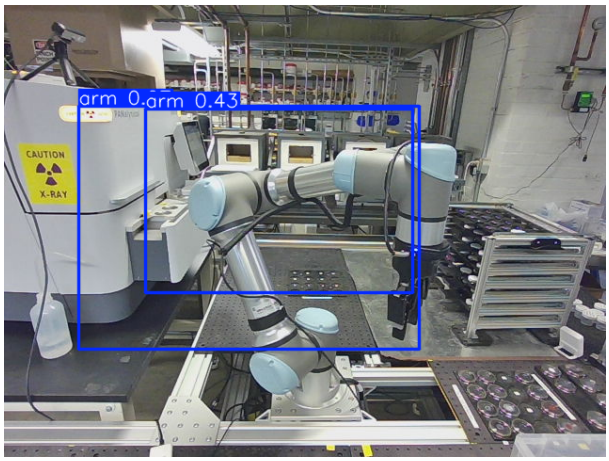
Detector output on a few keeps — blue box + confidence if present.



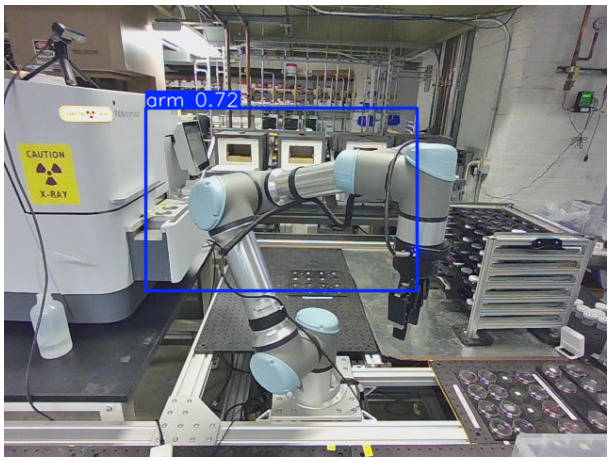
smoke_cam0_cam0_20261015_175717_seq00001.png



smoke_cam0_cam0_20261015_175717_seq00007.png



smoke_cam0_cam0_20261015_175717_seq00009.png



smoke_cam0_cam0_20261015_175717_seq00013.png

Full dataset on disk: `C:\Users\Ceder-ALAB\Desktop\Jessica\Jessica\robotics\arm_state\` (keeps/, labels/, dataset/, captures/).